

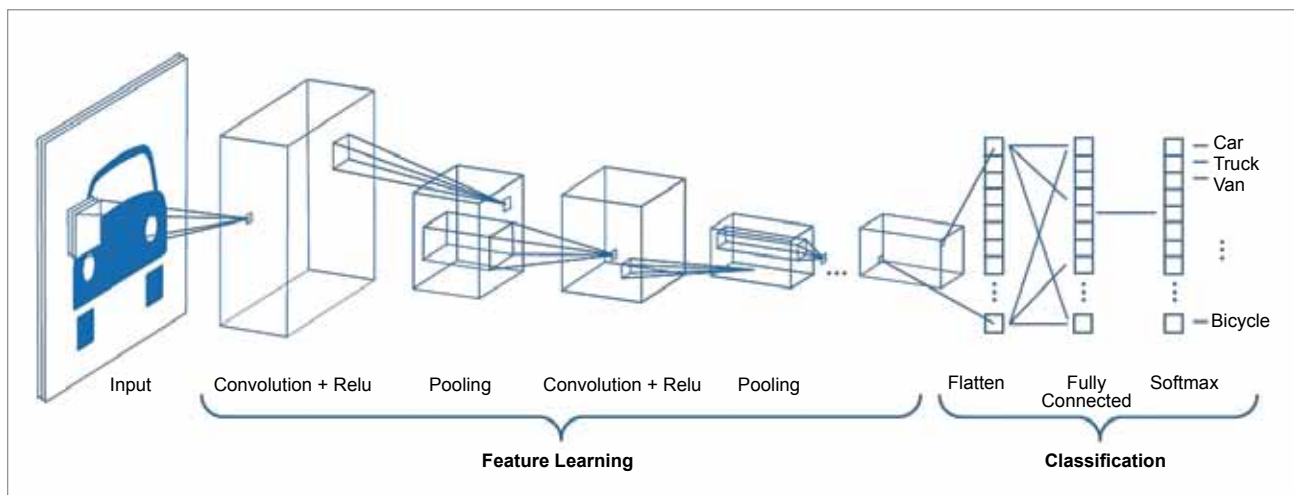


Fig. 1: Basic CNN architecture

LSTM networks are a variant of RNNs that are capable of learning long-term dependencies. The key to LSTMs is the cell state which stores information. The ability to add or remove information to the cell state is regulated by gates that are a sigmoid neural-network layer coupled with a point-wise multiplication operation. Since music sequences are time-series with long-term dependencies, it is appropriate to use LSTM.

CNNs are designed to process input images. Their architecture is composed of two main blocks. The first block functions as a feature-extractor that performs template-matching by applying convolution filtering operations. It returns feature maps that are normalised and/or resized. The second block constitutes the classification layer.

The network constitutes an input layer, hidden layers, and an output layer. In CNNs, the hidden layers perform convolutions. That includes typically a layer that does multiplication or dot product with rectified linear activation function (ReLU), followed by other convolution layers like the pooling layers and normalisation layers. A typical CNN architecture is shown in Fig. 1.

Generative adversarial networks (GANs) have components of generator, discriminator, and loss function. The generator role is to maximise the likelihood that the discriminator misclassifies its output as real. The discriminator's role is to optimise towards 50 per cent where the dis-

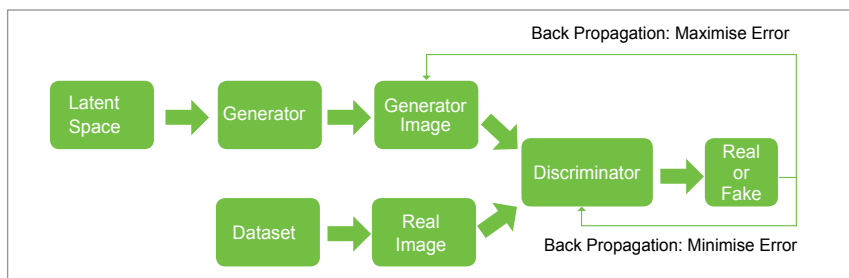


Fig. 2: Basic GAN architecture

criminator cannot identify between real and generated images. The generator will start training alongside the discriminator. The discriminator trains a few epochs prior to starting the adversarial training as the discriminator will be required to actually classify images. The loss function provides the stopping criteria for the generator and discriminator training processes.

Fig. 2 through Fig. 4 show pictorial block diagrams of the GAN architecture, the GAN generator, and the GAN discriminator. GANs are better than LSTMs in producing music because these are able to capture the large structural patterns in music.

Magenta: A Google Brain project

Magenta was an open source project developed by Google Brain in 2016 for the purpose of creating a new tool for artists when they work on developing new songs or art-work. Magenta is powered by Google's TensorFlow machine learning platform.

Magenta can work with music and images. In the music domain, an agent

automatically composes background music in real time using the emotional state of the environment in which it is embedded. Beneath, it chooses an appropriate composition algorithm dynamically from a database of previously chosen algorithms mapped to a given emotional state.

Doug Eck and team worked with an LSTM model tuned with reinforcement learning. Reinforcement learning was used to teach the model to use certain rules while still allowing it to retain information learnt from data.

The LSTM model works on a couple of metrics—the metrics we want to be low and the metrics we want to be high. The metrics associated with penalties are notes-not-in-key, mean-autocorrelation (since the goal is to encourage variety, the model is penalised if the composition is highly correlated with itself), and excessively-related-notes (LSTM is prone to repeating patterns). Reinforcement learning is brought in for creativity.

The metrics associated with rewards are compositions starting